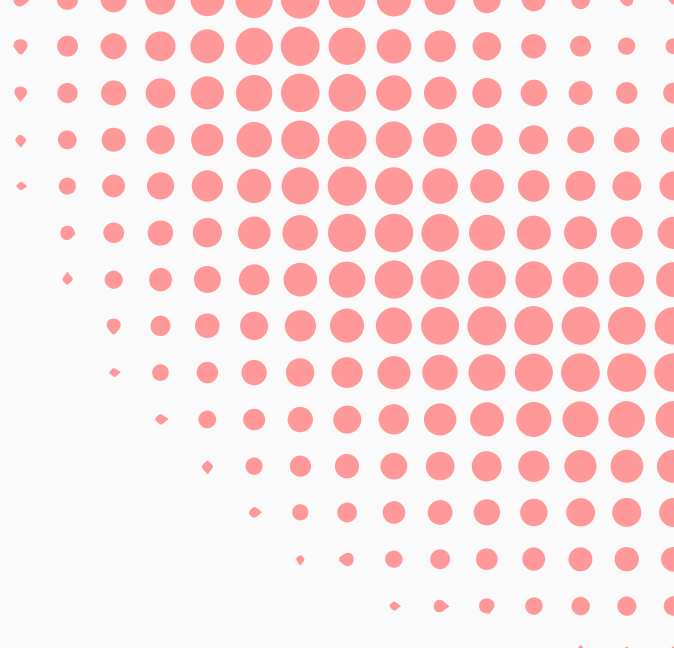


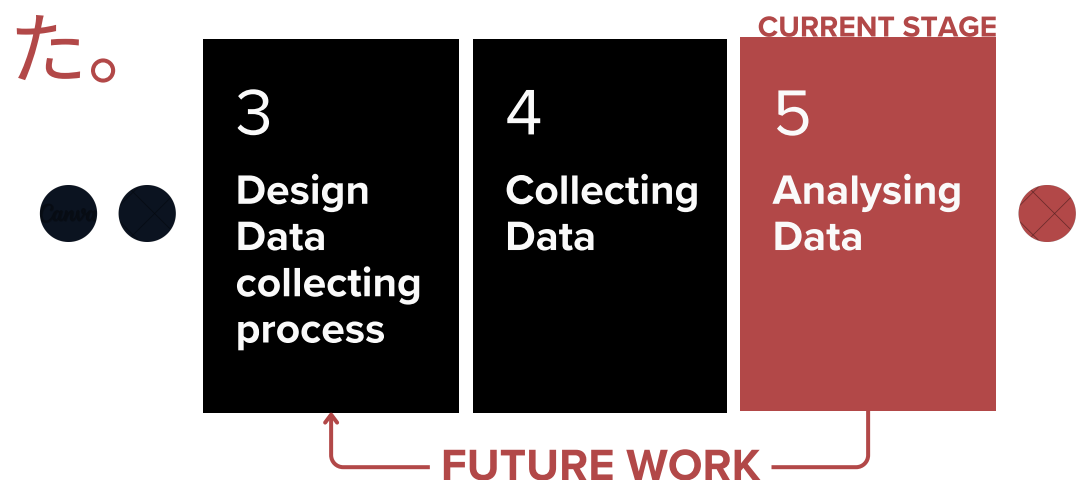


ANALYSIS OF HUMAN REACTION TO DIFFERENT MUSCLE STIFFNESS AND APPLICATION TO THE ROBOT

by Ben Srisoontorn

This week's progress

- **Recollect data (trying to apply constant force)**
データを収集する（一定の力を加えながら）
- **Plot and analyse the relationship between the measured and calculated force**
測定された力と計算された力の関係をプロットし、分析しました
- **Solve the cross talk problem of Arduino**
Arduinoにおけるクロストークの問題を解決しました
- **Complete the circuit to see difference between the different sensors**
異なるセンサー間の違いを確認するために回路を完成させました。



データの再収集

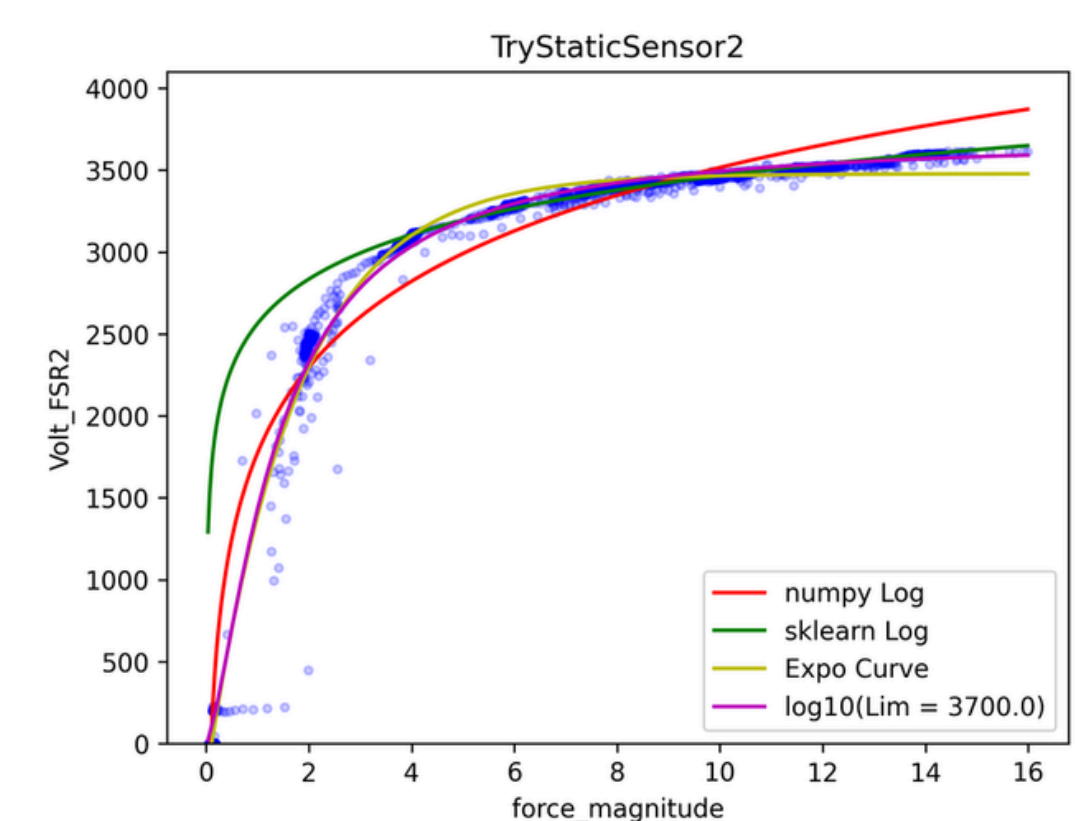
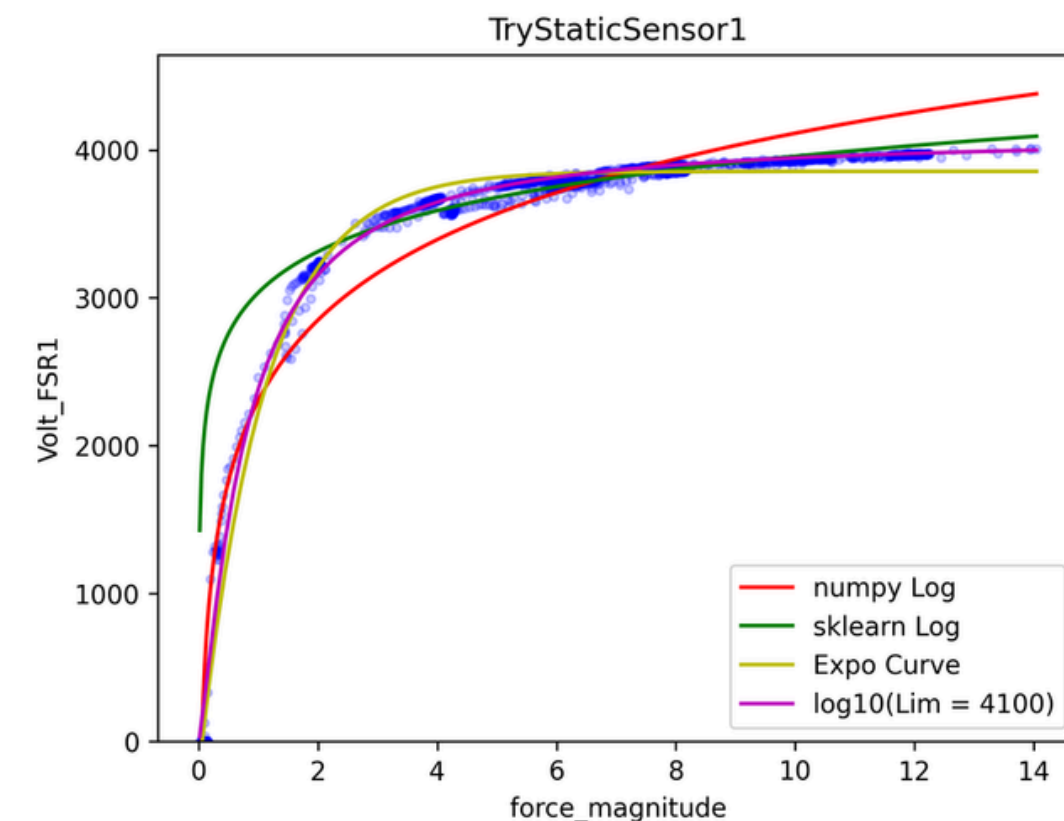
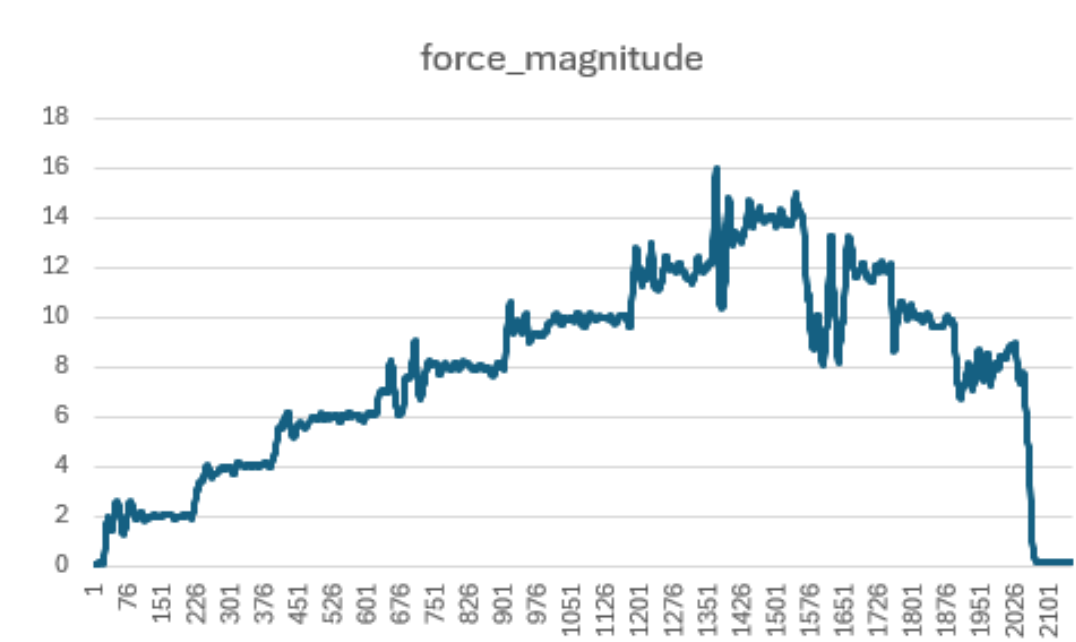
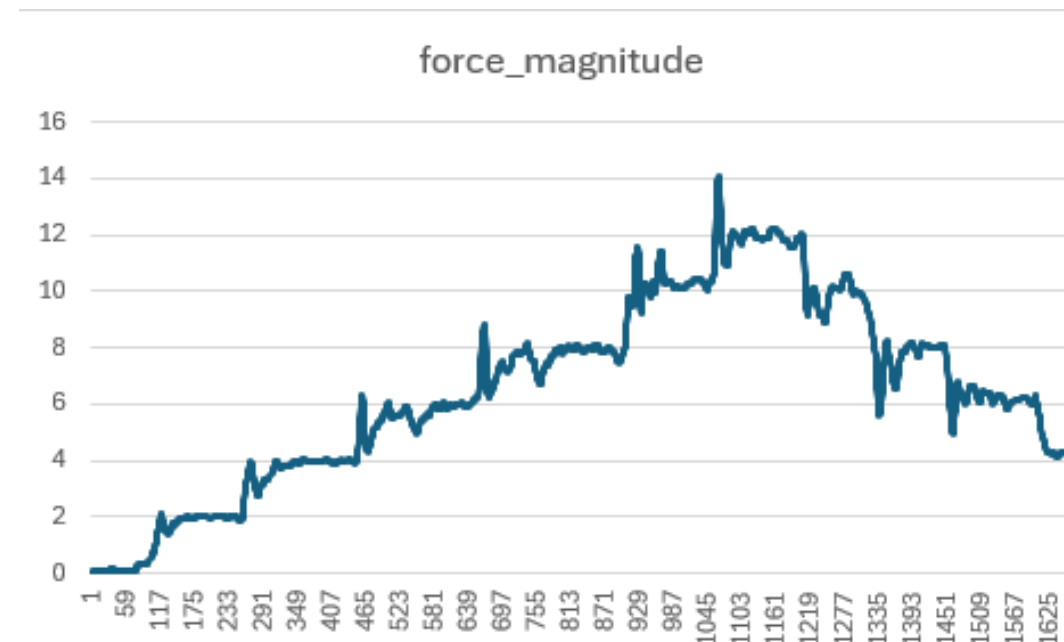
Recollect Data

Try to apply static force

静的な力を加えてみてください

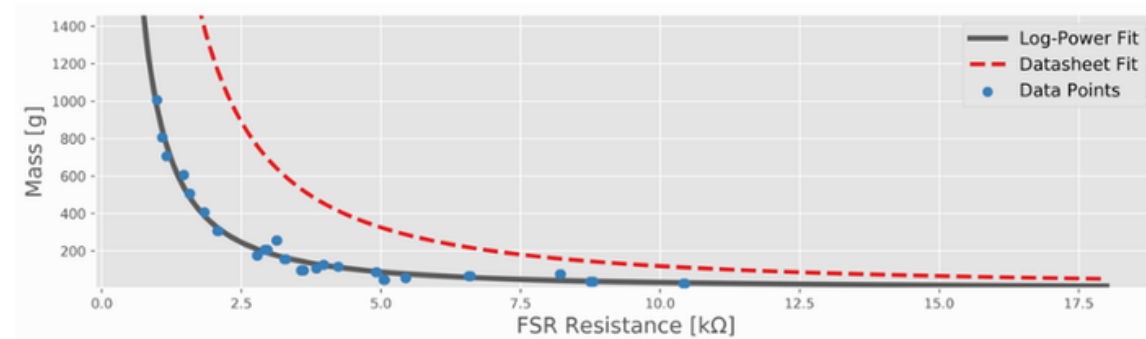
Collect data **same as before**
but try to **control the force** to
be in a range **during short**
period of time

以前と同様にデータを収集しましたが、今回は短時間、力を一定の範囲内に制御するように試みました



Measured and Calculated Force Comparison

From Article: Force Sensitive Resistors (FSRs) with Arduino



Resistance of FSR and applied force is in logarithmic relationship.

$$\log_{10}(m_i) = a \cdot \log_{10}(R_{FSR}) + b$$

Find coefficient to fit the curve of the Data

データの曲線に適合する係数を求める

$$\log_{10}(\text{CalculatedForce}) = \frac{1}{a} \cdot \log_{10}\left(\frac{V_{max}}{\text{ReadVoltage}} - 1\right) - \frac{b}{a}$$

FSR 1

$$a = -1.2677811420919507$$

$$b = -0.14258940729211392$$

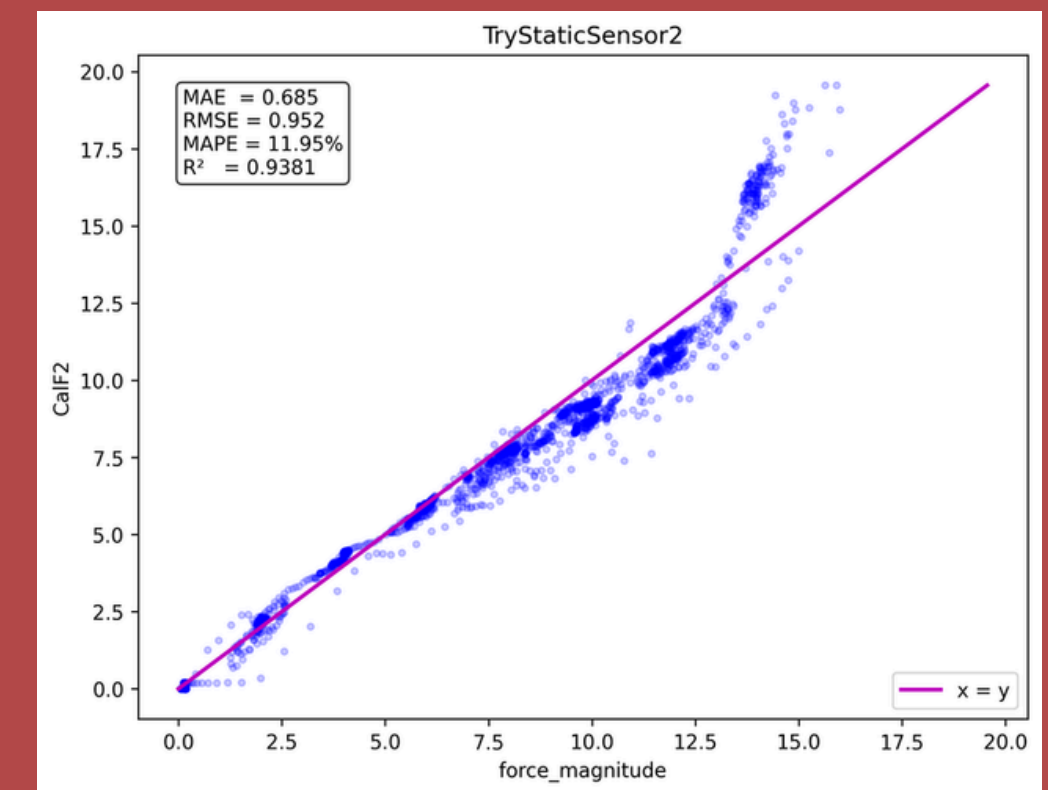
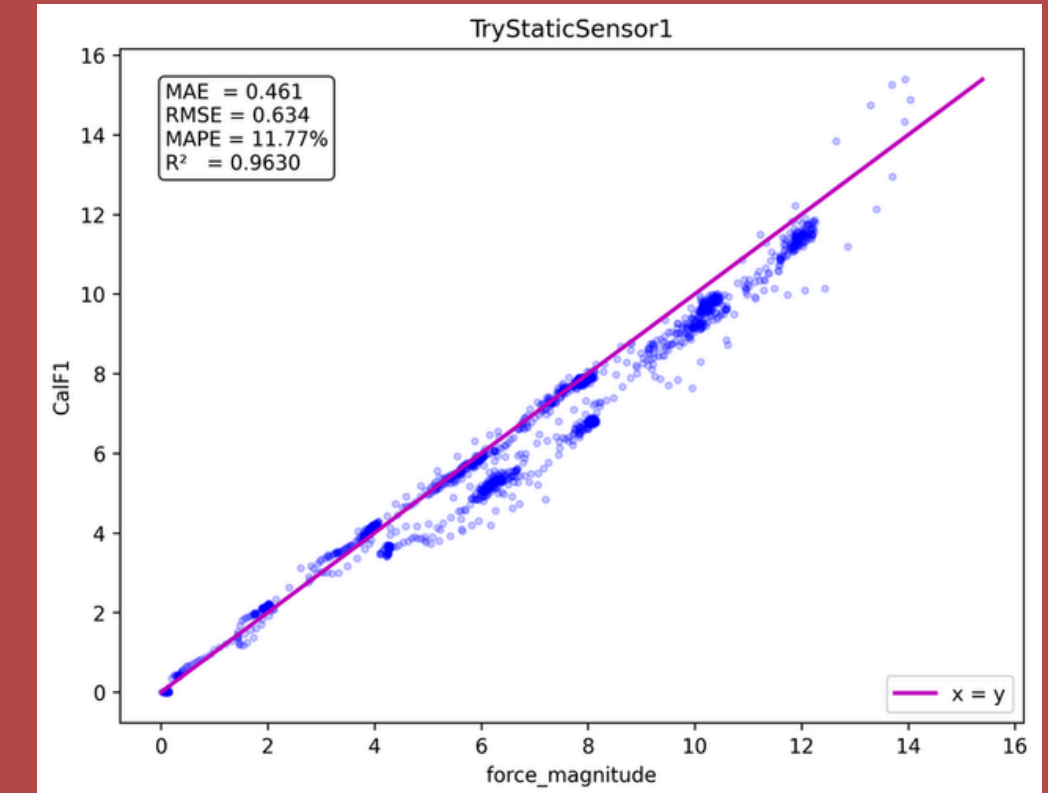
$$V_{max} = 4100$$

FSR 2

$$a = -1.4307043411643339$$

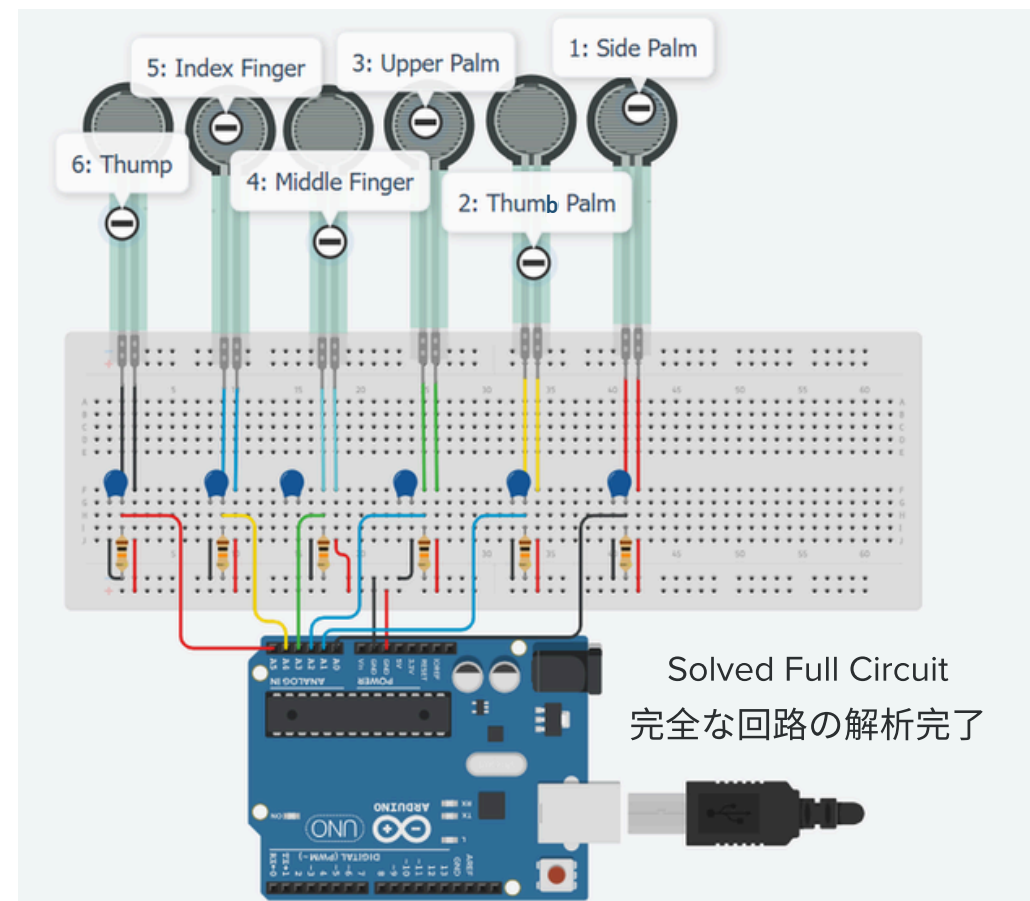
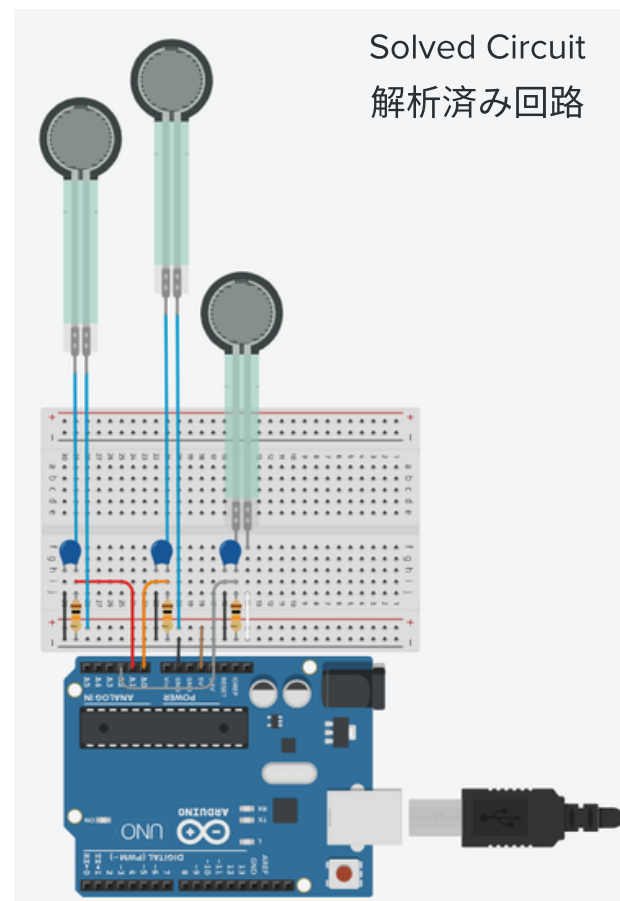
$$b = 0.20129777551613423$$

$$V_{max} = 3700$$

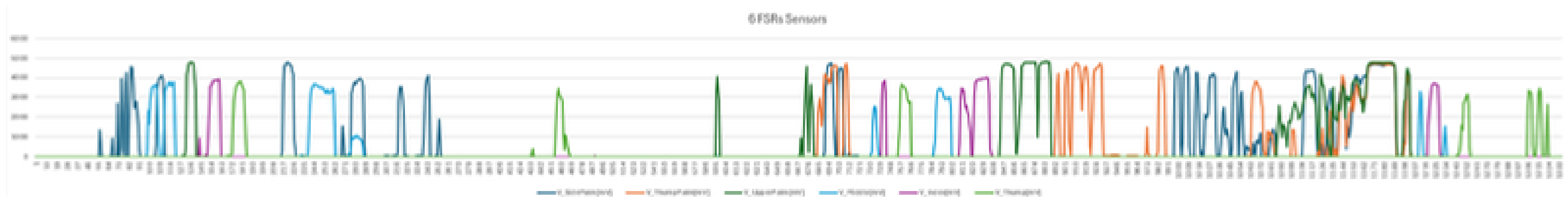


クロストーク問題の解決

Solving Crosstalk Problem



- Try ghost reading (reading the input 2 times) → not work
ゴースト・リーディング（入力を2回読み取る）を試す → 効果なし
- Read Gnd as reference → not work
GNDを基準電位（リファレンス）として読み取る → 効果なし
- Check wire and connection → checked nothing wrong
配線と接続を確認 → 異常なし
- Add 0.1 microF capacitor → working but some ignorable noise
< 3 mV read value
0.1 μ Fのコンデンサを追加 → 動作するが、無視できる程度のノイズ（読み取り値で3mV未満）あり



Result of 6 FSR together after Solved (6つのFSRを統合した結果（解決後）)

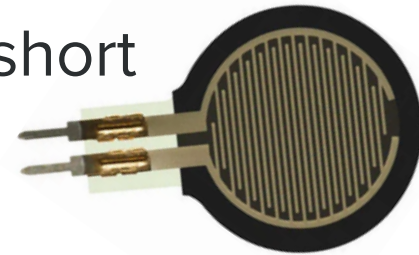
Collect Data

Recollect Data of 6 FSRs for Calibration and Testing

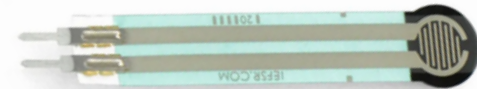
キャリブレーションおよびテスト用に、6つのFSRのデータを取得する

Included Sensor

Sensor 1, 2 and 3: FSR 402 short



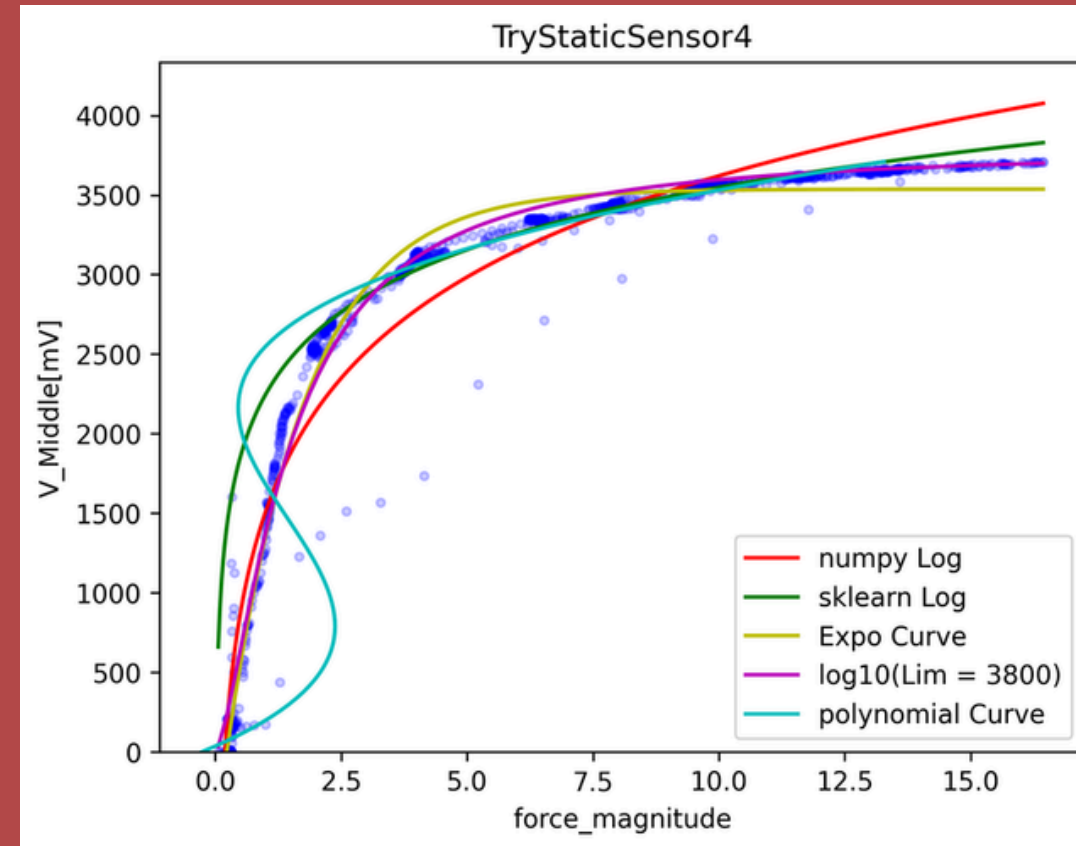
Sensor 4 and 5: FSR 400 long
(used for previous calibration test)
(以前の校正試験に使用)



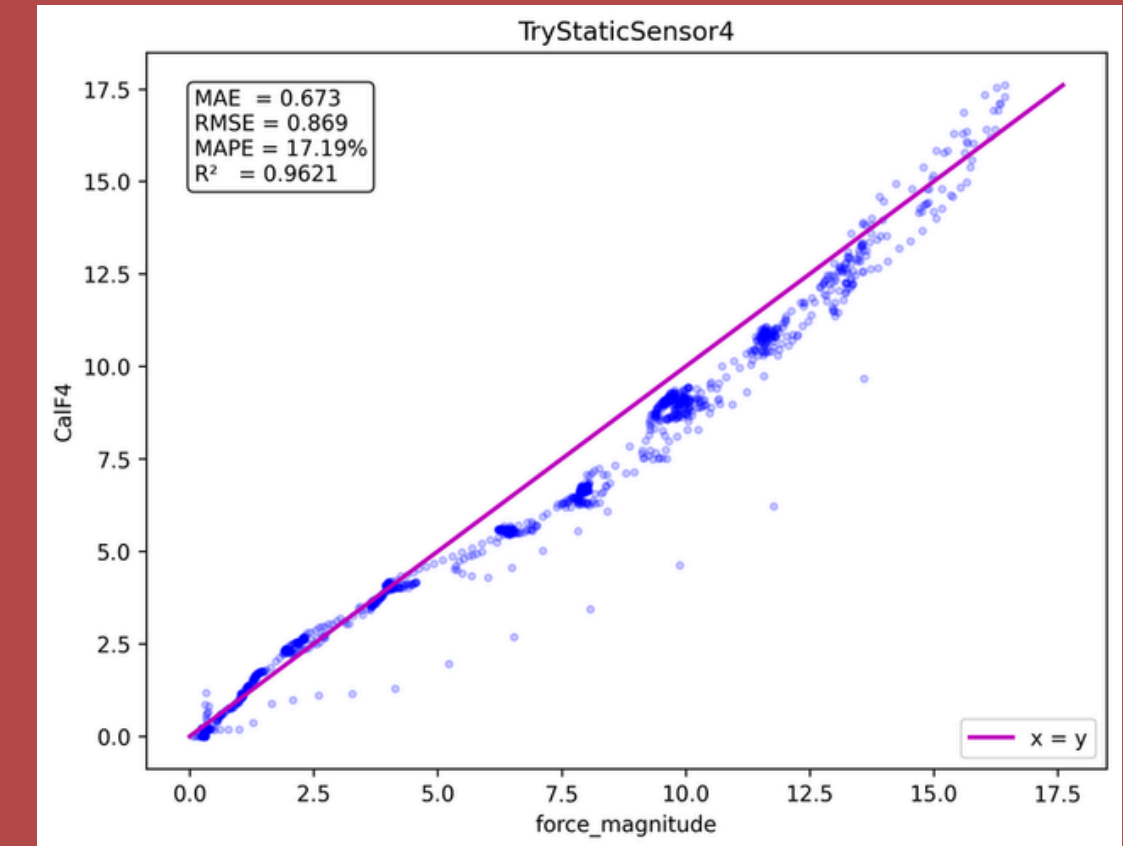
Sensor 6: FSR 400 short



Sensor4 for Middle Finger (Best outcome)



Analyse Data



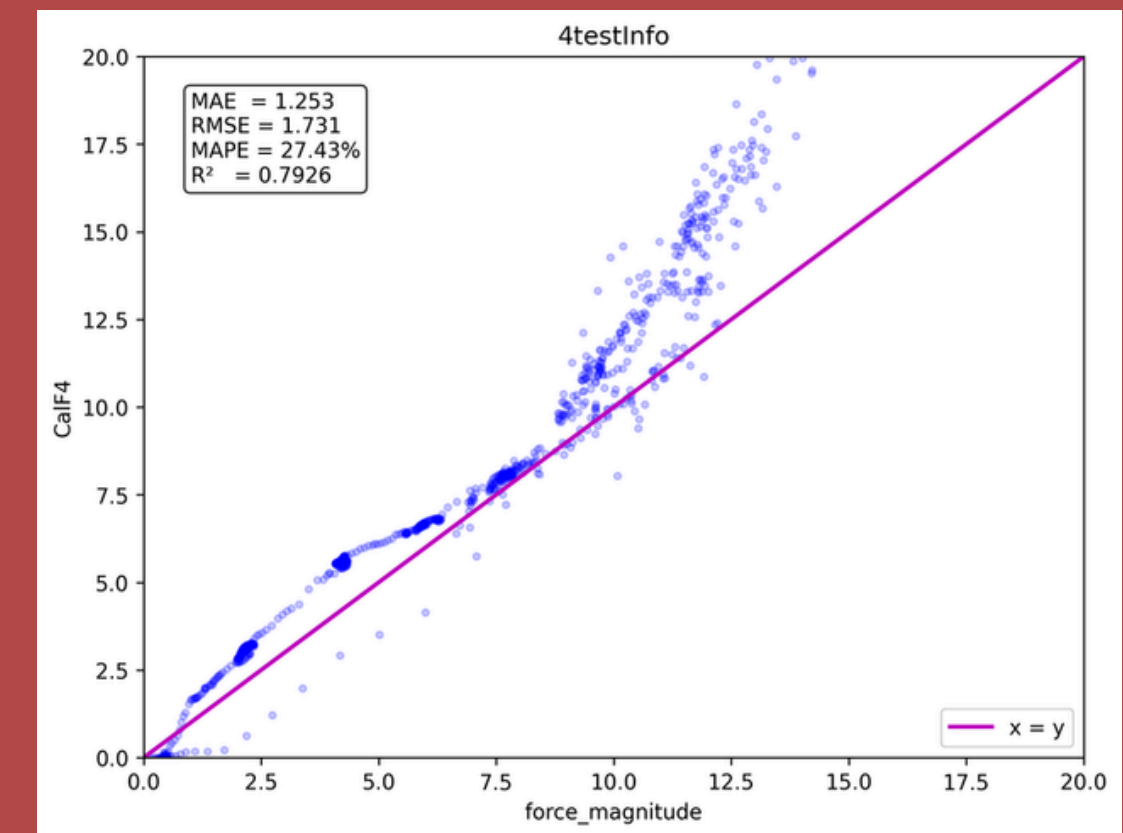
Analyse the result

Do regress → Find the curve fit
function → Plot CalF-MeasF for both
data for calibration and test data

回帰分析を行う → 曲線あてはめ関数を求め
る → 校正用データとテストデータの双方に
ついて、CalFとMeasFをプロットする

Result

- 0 - 5 N → error < 1N
- 5 - 10N → error < 2N
- > 10N → still not good and some cannot be calculated



Future Work

Main Research Topic:

主な研究課題:

- **Adjust the circuit and calibrate again for higher load FSRs (For Palm and Thumb)**

高負荷用FSR（手のひら・親指用）に合わせて回路を調整し、再校正を行う

Thai Massage for Open Campus:

オープンキャンパス向けThai Massage:

- **Sum up Thai Massage technique**

Thai Massage の技法の概要